

Transition Year · Student Book · Pilot edition

AI in Practice

Use it. Test it. Question it. Build with it.

A 30-hour, hands-on programme in applied artificial intelligence, digital skills and real-world problem solving.

Name: _____

Class: _____ School: _____

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Contents

Eight missions, thirty hours. Chapters 1–5 build the core ideas; 6 and 7 take them into study, work and the future; Chapter 8 is your project, where everything comes together.

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The badges you'll earn

Each chapter has a badge. Your teacher will confirm it when your portfolio evidence for that chapter is complete. Collect all eight to finish the programme as an **AI Innovator**.



AI Explorer



Machine Trainer



Data Detective



Prompt Engineer



AI Investigator



Digital Collaborator



Future Thinker



AI Innovator

How to use this book

This isn't a book about AI that you read and then get tested on. It's a set of eight missions. In each one you'll use a real tool, try to break it, measure what happened and decide what you think. You'll spend most of your time doing, not listening.

Every mission follows the same rhythm



The learning loop. You'll see it in every chapter.

You'll **predict** what will happen before you try anything, because being wrong on purpose is the fastest way to learn. You'll **test** rather than trust. And at the end you'll keep **evidence**: a map, a log, a rewrite, a prototype. That evidence builds up into your portfolio across the 30 hours.

What the boxes mean

🚩 Your mission

The one thing you must produce in each chapter. Everything else builds towards it.

🔍 Discover

A short explanation of an idea you need before you try it. Read it, then go and test it.

🔍 Key words

Words you'll hear and use in the chapter, with a plain meaning. The full glossary is at the back.

💡 Think about it

Questions with no single right answer. Use them in discussions and in your reflections.

📁 For your portfolio

The evidence to keep. Your portfolio, not a test, shows what you've learned.

★ Level up

An optional extra for when you want to push further. Never required.

☑ What you'll need

Tools and materials for the chapter. Your teacher will tell you which ones your school uses.

⚠ Safety rule

Things that keep you and other people safe when working with AI and data.

The one rule that runs through everything

AI outputs are guesses until you check them

Every answer an AI gives you is a *hypothesis*. It might be right. It might be confidently wrong. You are responsible for what you hand in, say or build, not the tool. When something matters, verify it against a source that doesn't come from an AI.

Keeping yourself and others safe

- Never type your own or anyone else's private details (full name, address, health, passwords, photos of people) into an AI tool. Use made-up examples instead.
- Only use tools your school has approved. If you're not sure, ask.
- You can always choose not to use AI for a task. That's a valid choice, and sometimes the better one.
- Some topics in this book (bias, misinformation, the future of work) can be uncomfortable. It's fine to say so, and to disagree with each other respectfully.

How you'll know you're doing well

There are no marks out of 100 here. Each chapter ends with a **How am I doing?** box with three levels: *Getting started*, *Getting there* and *Going further*. Be honest with yourself. Moving from one column to the next is the goal. You'll also earn a badge for each chapter you complete.

Mistakes and failed models count as evidence. A model that fails in an interesting way teaches you more than one that "just works".

Your portfolio

Throughout the book, the **For your portfolio** box lists what to keep. Blank sheets for the most common pieces of evidence are at the back (Portfolio sheets). Every piece of evidence should show four things: what you predicted, what you tried, what actually happened, and what you'd change.

Chapter 1

AI & Me

Badge: AI Explorer

2 hours

Links to LO1, LO4

1

Before we get to any technical words, let's find the AI that's already in your day. Some of it is obvious. Most of it is hidden inside apps you use without thinking. By the end of this chapter you'll be able to spot it, explain roughly what it's doing, and start asking who it helps and who it might harm.

By the end of this chapter you will be able to...

- Name at least five believable uses of AI in everyday life.
- Explain, in your own words, the difference between a fixed rule and a system that learns or generates from patterns.
- Give one benefit and one possible limitation of an everyday AI system.
- Ask a good question about how an AI system affects people.

Key words

Algorithm – a set of steps or rules for getting from an input to an output

Prediction – a system's best guess about something it hasn't seen yet

Recommendation – an output that suggests what you might want next

Pattern – something that repeats in data often enough for a system to pick it up

Automation – a task done by a machine instead of a person; not always AI

AI system – software that uses data and patterns to predict, decide or generate

Human oversight – a person checking, deciding or able to overrule the system

What you'll need

- A laptop, Chromebook or tablet, one per student or pair
- Sticky notes or a shared digital board
- Mission Sheet 1 (Portfolio sheet A1 works too)
- Optional: an approved AI chatbot for the comparison activity

🚩 Your mission

Create a one-page **AI in My Day** map showing at least five systems that use AI. For two of them, show the full chain: **INPUT** → **AI ACTION** → **OUTPUT** → **BENEFIT** → **RISK**.

The route

0–10 min

Hook: AI or not?

You'll see eight things: a calculator, a spam filter, a playlist recommendation, traffic lights, face unlock, autocorrect, a chatbot, a timer. Vote for each one: **AI**, **not AI** or **unsure**. Do this before anyone gives you a definition. Your gut answer is data too.

10–25 min

Discover: three questions that help

There's no perfect test, but three questions get you a long way.

1 Does it use data?

Not just a fixed setting or a timer, but information about things, people or the past.

2 Does it find patterns?

Does it recognise or generate patterns rather than follow one rule written by a person?

3 Does it adapt or predict?

Could it give a different answer tomorrow, based on what it has seen, without being reprogrammed?

Go back to the eight examples. Some will be clear (a timer fails all three). Some are borderline, and that's fine. Don't force every example into one box.

25–50 min

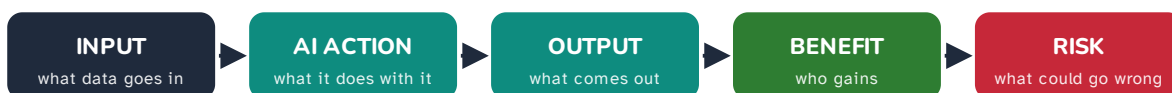
AI in My Day map

Draw a line for a typical day, waking up to going to sleep. Along it, mark every point where an algorithm might be influencing a *choice*, a *ranking*, a *prediction* or something *generated* for you. Think: alarm, phone unlock, messages, music, maps, social feeds, games, shopping, homework tools, streaming.

50–70 min

The influence lens

Pick three systems from your map. For each one, fill in the chain below.



Example: your listening history → finds patterns in what similar people play → a playlist → you find new music → you only ever hear more of the same

Add one more question: **who might be affected?** Not just you. The person who didn't get recommended. The shop that didn't appear on the map.

70–95 min

Human vs AI

In a small group, answer two questions: *What are humans better at?* and *What are current AI systems better at?* If you have an approved chatbot, ask it the same two questions. Then criticise **both** lists. Where does the AI's list flatter itself? Where does yours?

95–115 min

Micro-debate

The statement: **“Recommendation algorithms mostly help us.”** Stand on a line from *agree* to *disagree*, with *unsure* in the middle. Give a reason backed by an example from your map. You're allowed, and encouraged, to move after hearing other people.

115–120 min

Exit reflection

Write two sentences: *One thing I now notice about AI...* and *One question I want answered later...* Keep the question. We'll come back to it.

**Think about it**

- What evidence makes you think this thing contains AI, rather than just a rule someone wrote?
- Who decides what a “good” output is? You, the company, or the system?
- If the system gets it wrong, who notices? How quickly?
- When something is convenient, do we hand it more influence than we mean to?

Myth-busters
People say

“AI means robots.”

Actually

Most AI is invisible software: a ranking, a filter, a suggestion. Robots are a small corner of it, and many robots contain no AI at all.

People say

“Every computer program is AI.”

Actually

A timer, a calculator and most traffic lights follow fixed rules a person wrote. AI systems learn patterns from data or generate new output. Run the three questions.

People say

“If a system uses AI, it must be objective.”

Actually

AI learns from data made by people, chosen by people, about people. It can be very consistent and still be very unfair.

 For your portfolio

- ☐ Your AI in My Day map
- ☐ One human-vs-AI comparison, with your criticism of both lists
- ☐ Your exit reflection: one thing you notice, one question you want answered

 How am I doing? Tick what you can honestly say.**Getting started**

- ☐ I can point to examples of AI in everyday life
- ☐ I can say one benefit and one risk of an example

Getting there

- ☐ I can explain input, AI action and output for my examples
- ☐ I can question one limitation of a system

Going further

- ☐ I can compare systems, or human and AI abilities
- ☐ I can talk about influence, who's affected, or feedback loops

 Level up (optional challenge)

Choose one recommendation system (music, video, shopping). Sketch a **feedback loop**: how do your clicks today change what you get shown tomorrow, and how does that change what you click next?

Badge earned: AI Explorer ✓

Chapter 2

Teaching a Machine

Badge: Machine Trainer

4 hours

Links to LO1, LO2, LO3, LO9

You're going to teach a computer to tell two objects apart, then do everything you can to confuse it. Machine learning is an experiment, not magic. You'll create labelled examples, train a small model, test it on things it has never seen, count the mistakes, and then discover how a model can cheat without anyone telling it to.

By the end of this chapter you will be able to...

- Explain labels, features, training data, test data and prediction.
- Train a basic classifier using an approved visual tool or a simulator your teacher provides.
- Record predictions and work out simple accuracy.
- Read a basic confusion matrix.
- Explain how training data that doesn't represent the real world creates poor or biased results.

Key words

Supervised learning – teaching a model with examples that already have the right answer attached

Class label – the category an example belongs to, e.g. “cup” or “bottle”

Feature – something measurable about an example: colour, shape, edges, brightness

Training data – the examples the model learns from

Test data – new examples used to check the model; should not be reused from training

Model – the trained system that makes predictions

Prediction – the model's guess for a new example

Accuracy – $\text{correct predictions} \div \text{total predictions}$

Confusion matrix – a table showing which classes got mixed up with which

Generalisation – how well the model handles examples it wasn't trained on

Shortcut learning – when the model relies on an easy, unintended clue (like background colour) instead of the real thing

Bias – a systematic skew that makes results unfair or unrepresentative

☑ What you'll need

- A visual classifier tool such as Teachable Machine, or an equivalent your school approves
- Two safe, everyday objects per group (cup/bottle, pen/eraser...)
- A device with a camera if your school allows it; otherwise your teacher will give you images
- A test log (Portfolio sheet A3)
- A calculator or spreadsheet (optional)

🚩 Your mission

Train a two-class model. Produce a test log with at least 10 rows, build a simple confusion matrix, find one failure pattern, and improve the model. Then show the before-and-after evidence.

The route

0–20 min

Be the classifier

Write a set of rules a stranger could follow to tell your two objects apart. Swap rules with another group. Their job: find unusual examples that break your rules. A cup with no handle? A bottle lying on its side? Notice how fast written rules run out.

20–40 min

Discover: from rules to learning

Instead of writing every rule, machine learning gives the model lots of **examples** with **labels** and lets it find the **features** that separate them. It learns statistical patterns. Nobody types “cups have handles”.



The test set must be genuinely different from the training set, or you're only testing memory.

40–75 min

Build model v1

Collect a balanced set of examples for each class, roughly the same number of each. Train. Write down how many examples you used and the conditions: lighting, background, angles. You'll need this later.

75–100
min**Break your model**

Now test it on things it hasn't seen: a different cup, a new angle, weird lighting, a cluttered background, the object half hidden. **Your goal is to find failures, not to get a high score.** A model that never fails in your tests just means your tests are too easy.

100–120
min**Evidence checkpoint**

Record at least 10 tests in your log: the *actual* class, the *predicted* class, and correct or incorrect. Then work out accuracy: $\text{correct} \div \text{total}$. Eight out of ten is 80%.

120–145
min**Discover: the confusion matrix**

Accuracy hides *which* mistakes you made. Put the same 10 results into a 2x2 grid.

What the model predicted

	Cup	Bottle
was really		
Cup	Correct cup → cup	Missed cup called a bottle
Bottle	False alarm bottle called a cup	Correct bottle → bottle

Two kinds of mistake. Which one matters more depends on the job.

Think about a spam filter versus a medical screening test. A spam filter that lets a few spam emails through is annoying. A screening test that misses real illness is dangerous. Same table, very different stakes. No advanced maths needed, just ask: which error hurts more here?

145–175
min**The shortcut experiment**

Now deliberately build a flawed dataset. Put almost all of Class A on a **light** background and almost all of Class B on a **dark** background. Train. Then test with the backgrounds swapped. Did the model learn the object, or did it learn the background? This is **shortcut learning**, and it is how real systems end up biased without anyone intending it.

175–210
min**Improve: model v2**

Propose and apply at least two changes: more variety, better balance, more examples, mixed backgrounds, clearer labels. Retrain. Test using **the same test set as v1** wherever possible, so your comparison is fair.

210–230
min**Compare v1 and v2**

Present to another group: what you changed, what improved, and what is still uncertain. Numbers plus examples. If v2 is better on your tests but you're not sure why, say so.

230–240
min

Reflection

Write: *Why might “high accuracy” on its own not mean a model can be trusted?*
Use your own shortcut result as evidence.

💡 Think about it

- What pattern might your model be using that you didn't intend?
- Is your test set genuinely different from your training set?
- Which mistakes matter most for the job this model would do?
- How do you know an improvement is real and not just luck?

🚫 Myth-busters

People say

“The model sees the object the way I do.”

Actually

It sees pixels and statistics. It doesn't know what a cup *is*. It knows which patterns of pixels came with the label “cup”.

People say

“More examples automatically means better data.”

Actually

A thousand photos of cups all on the same white table teaches the model about white tables. Variety and balance beat volume.

People say

“Accuracy tells the whole story.”

Actually

Two models can have the same accuracy and make completely different mistakes. The confusion matrix shows which ones.

People say

“Bias only happens when someone codes unfair rules on purpose.”

Actually

Your background experiment proved otherwise. Bias creeps in through data nobody looked at closely.

📁 For your portfolio

- ☐ Your dataset plan: classes, number of examples, conditions
- ☐ v1 test log and accuracy (Sheet A3)
- ☐ Your confusion matrix
- ☐ The shortcut-learning observation: what you did and what happened
- ☐ Your v2 improvement note: what you changed and why
- ☐ Your reflection

✓ How am I doing? Tick what you can honestly say.

Getting started

- ☐ I trained and tested a model, with help
- ☐ I recorded some results

Getting there

- ☐ I tested on unseen examples and worked out accuracy
- ☐ I explained an error pattern and improved the model

Going further

- ☐ I can read a confusion matrix
- ☐ I spotted shortcut learning or bias
- ☐ I can justify how I chose to evaluate

☆ **Level up (optional challenge)**

Work out **precision** or **recall** for one class from your confusion matrix. Then explain, in plain words, a situation where that number matters more than overall accuracy.

Badge earned: Machine Trainer ✓

Chapter 3

Data Detective

Badge: Data Detective

3 hours

Links to LO5, LO6, LO9

3

3

Data is the raw material of AI. Every model in this book, and every app on your phone, runs on it. This chapter is an investigation: what does a service actually collect, what does it work out about you without you telling it, and who gets left out when a dataset is built? You'll finish by designing a better data plan than the one you started with.

By the end of this chapter you will be able to...

- Tell apart volunteered, observed and inferred data.
- Investigate how a digital service may collect and use data.
- Explain why purpose, minimisation and representativeness matter.
- Describe what GDPR and the EU AI Act mean for someone your age.
- Spot a dataset that could leave people out or misrepresent them.

🔍 Key words

Personal data – any information about a person who could be identified from it

Volunteered data – information you give on purpose

Observed data – information recorded while you use something

Inferred data – information a system guesses about you from other data

Cookies – small files a website stores on your device to remember or track you

Tracking – following what you do across pages, apps or time

Web scraping – automatically copying data from public web pages

Public / private data – whether data is openly available or restricted; “public” doesn't mean “free to use for anything”

Data minimisation – collecting only what's genuinely needed for a stated purpose

Consent – agreeing to a use of your data, freely and with real understanding

GDPR – the EU's data protection law; gives people rights over their data

EU AI Act – an EU law that regulates AI by how risky its use is

Representativeness – whether a dataset actually reflects all the people it will affect

☑ What you'll need

- Data Detective worksheet (Sheet A5 is the responsible data card)
- Short extracts from privacy policies, chosen by your teacher
- A browser
- Scenario cards
- A spreadsheet (optional)

⚠ Investigate at category level only

You'll look at *what kinds* of data a service collects ("location", "contacts", "viewing history"). Never open, screenshot or share your own account data, and never anyone else's. The point is the pattern, not your private life.

🚩 Your mission

Audit one digital service at category level, then design a **Responsible Data Card** for it: what is collected, why, what is inferred, the risks, what controls users have, and what should *not* be collected at all.

The route

0–15 min

Data trail warm-up

Pick a music, video or maps service. List what it can know *directly* and what it could *work out*. Sort your list into three piles.

Volunteered

You typed it in or chose it.
Your name, a playlist name,
a "like", a search.

Observed

Recorded while you use it.
How long you watched, where
you were, what you skipped.

Inferred

Guessed from the other two.
Your mood, your age range,
what you'll buy, who you know.

15–40 min

Data Tracking Sherlock

Choose one familiar service. Using its privacy policy or app-store listing, find the *categories* of data it says it collects. Note anything surprising. Note anything vague ("to improve our services" covers a lot).

40–65 min

Why collect it?

For five data categories you found, complete the chain: **COLLECTION** → **PURPOSE** → **USER BENEFIT** → **POSSIBLE RISK**. Be fair: some collection is genuinely useful. Some isn't.

65–85 min

Cookie and scraping mini-lab

Your teacher will show how cookies and tracking work, and how web scraping copies data from a public page. Then discuss: if something is publicly visible, is every reuse of it reasonable? A photo posted for friends is public. Is training a face-recognition model on it fair?

85–110 min

Dataset fairness challenge

Your group gets a scenario: an AI system for *school club recommendations, job shortlisting or transport planning*. Decide what data you'd collect. Then ask the hard question: **who is missing?** Students who joined mid-year? People without smartphones? Those who work nights?

110–135 min

Discover: your rights and the safeguards

GDPR gives you rights. The principles that matter most for you: data should be collected for a clear **purpose**; only what's needed (**minimisation**); kept **accurate** and **secure**; used **transparently**; and you have **rights** to see it, correct it and sometimes delete it. The EU AI Act adds a second idea: the riskier the use of AI (say, deciding who gets a job), the stricter the rules. You don't need the legal detail. You need the instinct.

135–165 min

Design a better data plan

Go back to your scenario. Remove fields you don't need. Add a check for who's represented. Decide how long data is kept. Mark the decisions that must have a **human reviewing** them before anything happens to a person.

165–180 min

Reflection

“Just because data is available, does that mean it should be used?” Take a position and back it with one concrete example from today.

 Think about it

- Does the service really need this data to do its job?
- Could it achieve the same goal with less information?
- Who is missing from this dataset, and what happens to them?
- What could be inferred about you even if you never typed it in?

🚫 Myth-busters

People say

"If data is public, any use of it is ethical."

Actually

Public means visible. It doesn't mean the person agreed to every possible reuse. Context matters.

People say

"Anonymous data can never cause harm."

Actually

Combine a few "anonymous" facts (postcode, age, school) and you can often identify a person. And group-level data can still be used unfairly against a whole group.

People say

"Consent makes any data use OK."

Actually

Consent buried in 40 pages nobody reads isn't real consent. And even genuine consent doesn't make a harmful or unfair use acceptable.

📁 For your portfolio

- ☐ Your data category audit of one service
- ☐ Your Responsible Data Card (Sheet A5)
- ☐ Your dataset fairness redesign: before and after
- ☐ Your reflection

✅ How am I doing? Tick what you can honestly say.

Getting started

- ☐ I can name categories of data a service collects

Getting there

- ☐ I can tell collected data from inferred data
- ☐ I can explain purpose, risk and minimisation for my service

Going further

- ☐ I can evaluate who's represented and propose safeguards
- ☐ I can justify why some available data shouldn't be used

★ Level up (optional challenge)

Compare two services that do a similar job. Using only publicly available information, explain which one appears to take the more data-minimising approach, and what evidence you're relying on.

Badge earned: Data Detective ✓

Chapter 4

Generative AI & Prompting

Badge: Prompt Engineer

4 hours

Links to LO7, LO9

4

4

Chatbots and image generators are the AI most people mean when they say “AI”. This chapter teaches you to work with them as a thinking and creating tool while staying in charge of the goal, the checking and the final call. There are no magic words here. You'll learn prompting the way you learned model training: by experimenting and comparing.

By the end of this chapter you will be able to...

- Explain, at a high level, what a large language model does and doesn't do.
- Improve a prompt using context, task, constraints and format.
- Compare outputs systematically instead of accepting the first one.
- Use AI for explanation, ideas, critique and transformation.
- Verify important factual claims and spot hallucination risk.

Key words

Generative AI – AI that creates new text, images, audio or code from learned patterns

Large language model (LLM) – a generative model trained on huge amounts of text to predict and produce language

Pattern prediction – the LLM's basic move: predict what most likely comes next

Prompt – the instructions and context you give a generative AI

Context – background the model needs: who you are, what it's for

Constraint – a limit you set: length, level, what to avoid

Output format – the shape you want the answer in

Iteration – improving in rounds, comparing each version

Hallucination – a fluent, confident output that's wrong or made up

Verification – checking a claim against independent, reliable evidence

Source quality – how trustworthy a source is: who made it, why, and how it can be checked

What you'll need

- A school-approved generative AI tool, or a shared demonstration run by your teacher
- Prompt Lab worksheet (Sheet A4)
- A browser for fact-checking (Sheet A2)
- Scenario cards

Your mission

Run a **three-version prompt experiment**: v1 weak, v2 improved, v3 tested and revised. Keep the outputs. Then produce a **verification log** for at least three factual claims the AI made.

The route

0–20 min

Same task, different prompts

Every group asks the AI the same deliberately weak question. Compare what came back. What varies between groups? What's missing? What sounds confident but has nothing to back it up?

20–40 min

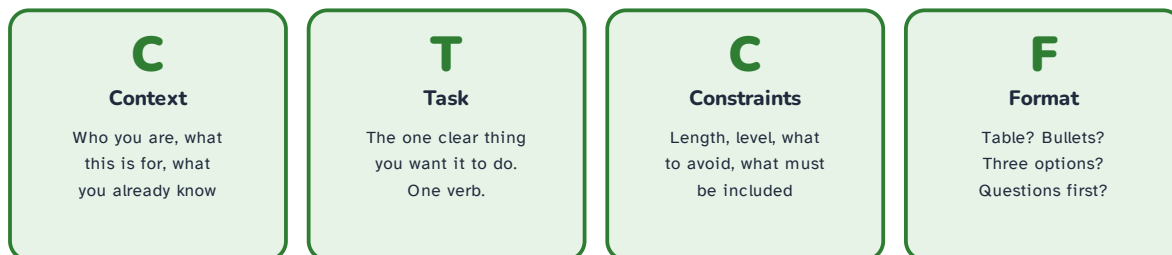
Discover: what an LLM actually does

An LLM has read an enormous amount of text and learned to predict what usually comes next. When you ask a question, it generates a likely continuation. That's why it's fluent. It's also why it's **not a database of checked facts**. It can produce a smooth, confident paragraph about something that never happened. **Confident wording is not confident truth**. Keep that sentence. You'll need it.

40–70 min

Prompt Lab 1: C-T-C-F

Take one weak prompt and rebuild it with four parts.



Save the before and after outputs side by side. That's your v1 and v2.

70–100 min

Prompt Lab 2: iterate

Now try adding one thing at a time: an example of what good looks like; the audience; success criteria; or “ask me questions before you answer”. Compare each version. Which additions actually helped, and which just made the prompt longer?

100–130 min

Four useful roles

Try the AI in four different jobs. Each has a different prompt style and a different risk.

- **Tutor:** “Explain X to me, then quiz me.” Risk: it explains something wrong, fluently.
- **Brainstorm partner:** “Give me 10 unusual angles on X.” Risk: you stop generating your own.
- **Critic:** “Here's my draft. What's weak?” Risk: it's too polite, or invents problems.
- **Transformer:** “Turn this into a table / a summary / simpler language.” Risk: it drops or changes meaning.

130–160 min

Verification challenge

You'll get a topic with several checkable facts. Ask the AI about it. Then pick three claims and verify each one yourself using reliable sources. Label each: **supported**, **uncertain** or **wrong**. Record it on Sheet A2.

160–185 min

Injecting doubt

Ask the model leading questions on purpose. “Why is X better than Y?” versus “Compare X and Y.” Watch how the framing of your question shapes the answer. This is **confirmation bias**, and the model will happily feed yours.

185–215
min**Build a reusable prompt**

Design a prompt template for a real TY task: preparing for work experience, learning a topic, planning an event, or making sense of survey feedback. Something you'd actually use again.

215–235
min**Peer red-team**

Swap prompts with a partner. Run theirs. Find one ambiguity, one missing constraint and one claim that would need checking. Give it back with notes.

235–240
min**Exit rule**

Finish this sentence: *“I should never trust an AI answer just because...”*

4

 Think about it

- What changed because of your prompt, and what changed just randomly between runs?
- Which parts of the answer need evidence from outside the AI?
- Are you asking the model to think *for* you, or to help you think?
- What would make this prompt safer or fairer for other people?

 Myth-busters**People say**

“Longer prompts are always better.”

Actually

Your Lab 2 comparison probably showed some additions did nothing. Clear beats long.

People say

“If it gives a citation, the source exists.”

Actually

LLMs generate citations the same way they generate everything else: by pattern. Some are real. Some are invented. Check.

People say

“Good prompt engineering guarantees a true answer.”

Actually

A better prompt gets a more useful answer. It doesn't turn a pattern-predictor into a fact-checker.

People say

“AI should give me the final answer.”

Actually

For learning, the answer is the least useful thing it can give you. The questions, examples and critique are worth more.

For your portfolio

- ☐ Prompts v1, v2, v3 with their outputs (Sheet A4)
- ☐ Your output comparison notes
- ☐ Your three-claim verification log (Sheet A2)
- ☐ Your reusable prompt template
- ☐ Your reflection and exit rule

How am I doing? Tick what you can honestly say.

Getting started

- ☐ I can write a usable prompt
- ☐ I notice when outputs differ

Getting there

- ☐ I improve prompts systematically
- ☐ I verify important factual claims

Going further

- ☐ I build criteria to judge outputs
- ☐ I compare outputs critically
- ☐ I can explain how framing and confirmation bias affect results

☆ Level up (optional challenge)

Write an evaluation rubric with four criteria. Score two AI outputs against it. Then justify which one is better *using the scores*, not just which one you prefer.

Badge earned: Prompt Engineer ✓

Chapter 5

Trust, Bias & Misinformation

Badge: AI Investigator

3 hours

Links to LO4, LO6, LO9

5

5

Fluent is not the same as true. In this chapter you'll learn to separate how something sounds from how reliable it is, hunt for the different ways bias sneaks into AI systems, verify claims the way a good journalist does, and then fix a piece of misleading AI-generated content so it's actually trustworthy.

By the end of this chapter you will be able to...

- Tell plausible-sounding language apart from verified evidence.
- Use lateral reading and source checks to investigate a claim.
- Identify examples of data, selection, framing and automation bias.
- Explain how generative AI can amplify misinformation.
- Produce a more trustworthy version of problematic AI-generated content.

Key words

Misinformation – false or misleading information, spread without meaning harm

Disinformation – false information spread on purpose to deceive

Hallucination – a fluent but wrong or invented AI output

Bias – a systematic skew in data, design or use that produces unfair results

Selection bias – when the data collected doesn't fairly represent the whole picture

Representation bias – when some groups are missing or under-counted in the data

Framing – how wording and emphasis shape what a message makes you feel or believe

Automation bias – trusting a machine's output more than you should, just because it's a machine

Synthetic media / deepfake – AI-generated audio, image or video that looks real

Provenance – where something came from and how it got to you

Lateral reading – checking a source by leaving it and seeing what other sources say about it

What you'll need

- Prepared claim cards
- A browser and search access
- AI News Detective worksheet (Sheet A2 for the verification table)
- An approved generative AI tool (optional)
- Three examples of sources of different quality

Your mission

Investigate one AI-generated article or set of claims. Produce an **evidence table**, identify at least one **bias risk**, and publish a **corrected version** with the uncertain parts clearly marked.

The route

0–15 min

The confidence trap

You'll see three confident statements. One is true, one is false, one can't be verified either way. Rank how confident you are in each *before* checking. Then check. Notice which one fooled you and why.

15–35 min

Discover: the trust checklist

Four habits, easy to remember, that work on news, on social posts and on AI output alike.

S**Stop**

Don't share or react yet. Notice how it makes you feel.

I**Investigate the source**

Who made this?
Why? What do others say about them?

F**Find better coverage**

Open new tabs. Do reliable, independent sources agree?

T**Trace the claim**

Follow it back to the original evidence. Was context lost?

35–70 min

AI News Detective

Generate, or receive from your teacher, a short news-style article written by AI. Go through it with a highlighter (real or digital). Mark: **factual claims**, **emotional framing**, **missing sources**, and **certainty that isn't backed up**. Count them.

70–100 min

Lateral verification

Pick 3–5 of the claims. Open new tabs and check each against independent, reliable sources. **Do not fact-check by asking another chatbot.** That's just asking a second pattern-predictor. Log each claim as supported, uncertain or wrong.

100–125 min

Bias stations

Four short scenarios, four stations: *hiring data*, *image generation and stereotypes*, *school discipline analytics*, *recommendation feeds*. At each one, ask: where could bias enter (the data? the design? how people use it?) and who would be affected?

125–150 min

Improve the output

Rewrite the article or a scenario response. Keep only the verified claims. Label anything uncertain as uncertain. Add real sources. Take out the loaded framing. Your version should be less exciting and more trustworthy. That's the point.

150–170 min

Synthetic media

Discuss voice, image and video generation. “Seeing is believing” was never fully true, and now it's weaker. The focus isn't on scary examples; it's on habits: what would you check before believing a clip that made you angry?

170–180 min

Reflection

Write your own **three-step rule** for checking important AI-generated information. Make it short enough to actually use.

💡 Think about it

- What would change your mind about this claim?
- Is this source independent of the original claim, or just repeating it?
- What's missing from the framing? What's not being said?
- Could a statement be accurate and still be misleading?

🚫 Myth-busters

People say

"Bias means the model has opinions."

Actually

Models don't have opinions. Bias is a skew in the data, the design or the use. It can exist with nobody meaning it.

People say

"Fake information looks unprofessional, so it's easy to spot."

Actually

Generative AI produces polished, well-formatted, confident text and images on demand. Looking professional now costs nothing.

People say

"Lots of websites say it, so lots of sources confirm it."

Actually

Ten sites copying one original claim is one source, not ten. Trace it back.

📁 For your portfolio

- ☐ Your annotated AI output
- ☐ Your verification table (Sheet A2)
- ☐ Your bias analysis from the stations
- ☐ Your corrected version
- ☐ Your personal three-step trust rule

✅ How am I doing? Tick what you can honestly say.

Getting started

- ☐ I can spot unsupported claims or possible bias, with prompts

Getting there

- ☐ I use independent sources to verify claims
- ☐ I can improve misleading content

Going further

- ☐ I trace evidence back to its origin
- ☐ I can tell different kinds of bias apart
- ☐ I can explain what's still uncertain after checking

 Level up (optional challenge)

Take one claim that's repeated widely online. Trace it back towards its earliest available source. Show where along the way context was lost or changed.

Badge earned: AI Investigator ✓

Chapter 6

AI for Learning & Work

6

Badge: Digital Collaborator

3 hours

Links to LO7

Everything so far has been about understanding AI. This chapter is about using it well: as a tutor that makes you think, an editor that doesn't take over, a planner and an analyst. The skill you're building is staying the author of your own work while still getting real help. That's what employers and teachers will want to see.

6

By the end of this chapter you will be able to...

- Use AI to support understanding, not just to produce answers.
- Choose the right AI role for a task.
- Apply a human-review checklist to AI-assisted work.
- Use spreadsheets, documents or presentations alongside AI on a small productivity task.
- Explain when AI use should be disclosed.

Key words

AI tutor – using AI to ask you questions and explain, rather than hand you answers

Scaffolding – support that helps you do something you couldn't do alone, then gets removed

Metacognition – thinking about your own thinking: what do I understand, and how do I know?

Drafting – producing a first version to improve, not a final one

Critique – structured, specific feedback on what's weak and why

Workflow – the sequence of steps a task goes through, including who or what does each one

Human-in-the-loop – a person checks, decides or approves at key points

Disclosure – saying clearly when and how AI helped

Authorship – being responsible for the work as yours

Digital productivity – getting useful things done with spreadsheets, documents and presentations

☑ What you'll need

- An approved AI tool
- A spreadsheet application
- A document or presentation tool
- Learning contract worksheet
- A messy mini-dataset from your teacher

🚩 Your mission

Complete a mini workflow that combines an **AI tutor task** with one **spreadsheet, document or presentation task**, and show the **human checks** you performed along the way.

6

The route

0–20 min

“Do it for me” vs “teach me”

Compare two conversations with the same AI: one that says “*Give me the answer*” and one that says “*Teach me through questions.*” Which one leaves more evidence that a person learned something? Which one would you rather defend to a teacher?

20–50 min

AI Tutor challenge

Pick a concept you genuinely want to understand, from any subject. Prompt the AI to: find out what you already know; give an example; ask you questions; and **hold back the full explanation until you've had a go**. Keep the conversation.

50–70 min

Explain it without AI

Close the tool. Explain the concept from memory, in writing or out loud to a partner. Compare with where you started. That gap is what you learned. If there's no gap, the tutor prompt needs work.

70–100 min

Workplace workflow

You'll get messy data: an event budget, survey results, costs. Clean it up in a spreadsheet yourself. Calculate a few values. *Then* ask the AI to suggest interpretations or wording for a summary.

100–125 min

Human review

Check the formulas and the data yourself. Remove any claim the AI made that your numbers don't support. Build one chart or one concise summary you'd be happy to put your name on.

125–150
min**Professional communication**

Draft a short professional email or briefing. Use AI as an **editor**, not as the author. Keep a note of what it suggested and what you changed or rejected. That note is evidence.

150–170
min**Disclosure scenarios**

In groups, decide whether AI use should be disclosed, and how, in each case: private brainstorming; assessed schoolwork; drafting a CV; a workplace report; a creative project. Expect disagreement. Work out what the disagreement is really about.

170–180
min**Your learning contract**

Write your own rules: “*AI helps me with...*” and “*AI does not replace...*” for schoolwork. Make them specific enough to catch yourself breaking them.

 Think about it

- What did you understand today that you couldn't explain before?
- Which part of this work are *you* personally accountable for?
- Which AI suggestion did you reject, and why?
- Would you be comfortable explaining your AI use to a teacher or an employer?

6

 Myth-busters**People say**

“Using AI efficiently means automating as much as possible.”

Actually

Automating the thinking removes the learning and the accountability. Efficient means automating the right parts.

People say

“If AI helped, the student didn't really learn.”

Actually

A student who used AI as a tutor, then explained the idea without it, has learned. The evidence is the explanation.

People say

“Disclosure is either always required or never required.”

Actually

It depends on the context and what people expect. Private brainstorming and an assessed essay are different situations.

For your portfolio

- ☐ Your AI tutor prompt and a note on what you learned
- ☐ Your spreadsheet, document or presentation
- ☐ Your human-review checklist, filled in
- ☐ Your AI-use learning contract

✓ How am I doing? Tick what you can honestly say.

Getting started

- ☐ I used AI for a learning or productivity task, with support

Getting there

- ☐ I showed my human checks
- ☐ I can explain what AI contributed and what I stayed responsible for

Going further

- ☐ I designed a repeatable workflow with human checkpoints
- ☐ I made disclosure choices and rejected some AI suggestions, with reasons

☆ Level up (optional challenge)

Design a repeatable AI-assisted workflow for a real TY activity (work experience prep, an event, a fundraiser). Mark every point where a human must **verify**, **decide** or **approve** before the workflow continues.

Badge earned: Digital Collaborator ✓

Chapter 7

Our AI Future

Badge: Future Thinker

3 hours

Links to LO4, LO8



Nobody knows what AI will look like in 2035, including the people building it. That's not a reason to stop thinking about it. In this chapter you'll build more than one possible future for a career or public service you care about, look at each one through the eyes of different people, and work out which outcomes depend on human choices rather than on the technology.

By the end of this chapter you will be able to...

- Tell the difference between today's narrow AI and the idea of AGI.
- Identify plausible opportunities and risks in a domain you choose.
- Build more than one future scenario instead of presenting one prediction as certain.
- Consider effects on jobs, skills, inclusion, sustainability and accountability.
- Communicate a reasoned position on a future AI choice.

7

Key words

Narrow AI – systems built for specific tasks: everything you've used so far in this book

AGI – artificial general intelligence: a hypothetical AI with broad, human-like capability across many tasks; not what exists today

Automation – a task done by a machine instead of a person

Augmentation – a person doing a task better or faster with a machine's help

Scenario – a plausible story about the future, built to think with, not to predict

Uncertainty – the honest gap between what we know and what we're guessing

Trade-off – gaining one thing at the cost of another

Accountability – who answers for a decision when it goes wrong

Future skills – human capabilities that become more valuable as AI spreads

Societal impact – effects on communities, fairness, work and wellbeing, not just on individuals

☑ What you'll need

- 2035 scenario canvas
- Career and domain cards
- Poster or presentation tool
- An approved AI tool for stress-testing your scenarios (optional)

🚩 Your mission

Create a **2035 scenario pack** for one career or public service. It must show three possible futures, the impact on different stakeholders, the future skills that matter, and one **governance choice** you'd argue for.

The route

0–20 min

2026 vs 2035

List the AI capabilities you can actually see now. Then list what you think might plausibly exist by 2035. Keep the two lists apart. One is observation. The other is prediction. Mixing them up is the most common mistake in this whole topic.

20–40 min

Discover: narrow AI and AGI

Every system you've used in this book is **narrow**: trained for a specific job. **AGI** is the idea of an AI with broad general ability across many kinds of tasks. It's a serious research goal, and also a word that gets used loosely in headlines. Timelines and impacts are genuinely uncertain, and experts disagree. When someone tells you exactly when it will arrive, ask how they know.

40–70 min

Career transformation map

Choose a career: teacher, doctor, farmer, architect, engineer, journalist, tradesperson, designer, accountant, or one of your own. Break it into tasks. For each task: likely to be **automated**, **augmented**, or **still strongly human**? Notice that a job is a bundle of tasks. Technology usually changes the bundle, not the whole job at once.

70–100
min**Three futures**

For the same domain, your group builds three scenarios.

Optimistic

Things go well.
What technology change?
What human response?
What unintended effect?

Concerning

Things go badly.
Same three questions.
Who carries the risk?

Balanced

Mixed, messy, likely.
Same three questions.
Which choices decided it?

Each scenario needs a technology change, a human response, and an unintended consequence. No scenario is allowed to be “everything’s fine” or “everything’s ruined”.

100–125
min**The stakeholder lens**

Go back through your scenarios as five different people: a *worker*, a *customer or citizen*, an *employer*, a *regulator*, and *someone at risk of being excluded*. Where do their interests clash?

125–150
min**Policy choice debate**

Pick a proposal, for example: “*AI-generated schoolwork must always be labelled*” or “*High-stakes AI decisions must always have human review.*” Prepare the arguments for it **and** the strongest arguments against it. Then debate.

150–170
min**Future skills card**

Identify three human capabilities that become *more* valuable in your chosen future, and one concrete thing you can do during TY to build one of them.

170–180
min**Reflection**

Which future outcome depends most on human choices, rather than on the technology itself?

 Think about it

- Which part of your scenario is evidence-based, and which is speculation?
- Who benefits? Who carries the risk?
- Does the technology replace a task, or a whole job?
- Which human capability becomes more important, not less?

🚫 Myth-busters

People say

“AGI already exists, in the same sense as today's chatbots.”

Actually

Today's systems are narrow, however impressive. AGI is a hypothesis and a goal, not a product you can download.

People say

“Every task that can be automated should be.”

Actually

“Can” and “should” are different questions. Some tasks are automated and worse for it. Your stakeholder lens showed you who decides.

People say

“Future-of-work predictions are facts.”

Actually

They're forecasts built on assumptions. Reasonable experts disagree. Treat them as scenarios.

📁 For your portfolio

- ☐ Your 2035 scenario canvas
- ☐ Your career transformation map
- ☐ Your stakeholder analysis
- ☐ Your future skills card
- ☐ Your debate reflection

7

✅ How am I doing? Tick what you can honestly say.

Getting started

- ☐ I can describe a plausible AI-related change in a career or domain

Getting there

- ☐ I built more than one scenario
- ☐ I identified trade-offs and future skills

Going further

- ☐ I separate evidence from speculation
- ☐ I analysed stakeholders
- ☐ I can defend a governance choice

☆ Level up (optional challenge)

Find two credible but different forecasts about AI and work. Explain why reasonable experts can disagree, and identify the assumptions behind each one.

Badge earned: Future Thinker ✓

Chapter 8

AI Innovation Project

8

Badge: AI Innovator

8 hours

Links to LO1–LO9

This is where the whole programme comes together. You'll pick a real problem in your school or community, understand the people who have it, design a responsible solution, build a rough prototype, test it on real people, improve it, and present it honestly, including what didn't work. AI is optional. Part of the project is deciding whether it genuinely helps.

By the end of this chapter you will be able to...

- Define a real user or community problem before choosing any technology.
- Research users and constraints without collecting unnecessary personal data.
- Design and prototype an AI-enabled or AI-informed solution.
- Test the prototype against clear success criteria.
- Identify limitations and risks: bias, misinformation, privacy and human oversight.
- Improve based on evidence.
- Present the solution, the evidence and what's still uncertain.

Key words

Problem framing – describing the problem clearly enough that you could tell if you'd solved it

User need – what the person with the problem actually needs, which may not be what they first ask for

Success criteria – how you'll know, before you build, whether it worked

Prototype – a rough, cheap version built to learn, not to impress

Testing – giving the prototype to real people and watching what happens

Iteration – changing it based on what you learned, then testing again

Risk – what could go wrong, for whom, and how badly

Responsible AI – designing so that harm is anticipated, limited and owned by a person

Human oversight – the decisions a person must keep

Value proposition – the one sentence that says why anyone would want this

Red-teaming – deliberately trying to break your own solution before someone else does

What you'll need

- Innovation Project Canvas
- Approved prototyping tools
- Optional: an AI or LLM tool, spreadsheet, slides, a no-code builder or a coding tool
- Peer test sheets
- The presentation guide at the end of this chapter

Your mission

Solve a real school or community problem with a **responsible, testable prototype**. AI is optional: your team must justify whether it adds genuine value, and be ready to say “it doesn't” if that's what your evidence shows.

Research safely

Only interview or survey people using questions your teacher has approved. Collect no personal data you don't need. Everything from Chapter 3 applies here, to your own project.

The route

Sprint 1 · hours 1–2
Understand

Sprint 2 · hours 3–4
Design

Sprint 3 · hours 5–6
Build & test

Sprint 4 · hours 7–8
Improve & show

Four sprints of two hours. You'll want to build in hour one. Don't. The problem comes first.

Hour 1

Sprint 1 · Problem hunt

Look at school and community life with fresh eyes. Where do people waste time, get confused, or get left out? Generate at least five problems. Choose one using five tests: **real, understandable, useful, testable, safe**.

Hour 2

Sprint 1 · Understand the user

Define the user, how they do things now, the pain points, the evidence, and the constraints. Run lightweight interviews or surveys, with approved questions only.

Hour 3

Sprint 2 · Design options

Generate **three** solution approaches, and at least one must not use AI. Compare them honestly. Decide whether AI genuinely adds value or just adds complexity.

Hour 4

Sprint 2 · Responsible design

Map the data flow. Mark the human decision points. List the failure modes and the safeguards. Then, **before you build anything**, write down your success criteria.

Hour 5

Sprint 3 · Build the prototype

The smallest version that tests your biggest assumption. A prompt workflow, a chatbot mock-up, a spreadsheet, a simple web page, a classifier, a simulated presentation, or code. Rough is fine. Rough is the point.

Hour 6

Sprint 3 · Test

Give it to classmates or real users. Record what worked, where they got confused, what failed, and **one thing you didn't expect**. Don't help them while they use it. Watch.

Hour 7

Sprint 4 · Improve and red-team

Make one change driven by test evidence. Then attack your own solution: could it hallucinate? Be biased? Leak private data? Be misused? Make people over-rely on it? Write down the limitations that remain. Honestly.

Hour 8

Sprint 4 · Demo and reflection

A 3–5 minute presentation using the structure below. Then complete your individual reflection.

Your presentation, in order (3–5 minutes)

1. The problem, and who experiences it.
2. The evidence you gathered.
3. The solution, and why AI is or isn't appropriate.
4. A demonstration of the prototype.
5. What happened during testing.
6. One change you made because of evidence.
7. One important risk, and its safeguard.
8. What's still uncertain, or what you'd test next.

Stuck for a problem? Some starting points

- A school FAQ or event assistant
- Study-planning or revision support
- A lost-property workflow
- Canteen or sustainability data analysis
- A local tourism information assistant
- Sports-club coordination
- A career exploration assistant
- Making community information more accessible
- A school survey insight tool
- A non-AI solution, because your team concluded AI adds no real value here

Think about it

- Would this problem still be worth solving without AI?
- What's the smallest prototype that can test your biggest assumption?
- What happens when the AI is wrong?
- Which decision must stay with a human?
- What evidence shows your change actually improved things?

🚫 Myth-busters

People say

"The project has to use the most advanced AI."

Actually

The project has to solve a problem. The best answer is the least complex one that works. Sometimes that's a spreadsheet.

People say

"A working demo proves the idea is useful."

Actually

A demo proves it runs. Test evidence from real users proves it's useful. They're different.

People say

"Hide the limitations in the presentation."

Actually

Naming the limitations is one of the things you're being assessed on. Teams that hide them look like teams that didn't test.

📁 For your portfolio

- ☐ Problem statement and user evidence
- ☐ Three solution options, including the non-AI one
- ☐ Responsible AI and data canvas
- ☐ The prototype
- ☐ Your test record
- ☐ Your iteration log
- ☐ Your risk register
- ☐ The final presentation
- ☐ Your individual reflection

✅ How am I doing? Tick what you can honestly say.

Getting started

- ☐ I defined a problem and built a basic prototype
- ☐ I can describe what it does

Getting there

- ☐ I used evidence from users
- ☐ I tested, improved and identified key risks

Going further

- ☐ I compared AI and non-AI options
- ☐ I set success criteria in advance
- ☐ I red-teamed the solution and communicated uncertainty honestly

How your project will be judged

Your teacher will use these six criteria. Read them before you start, not after.

	Getting started	Getting there	Going further
Problem	Problem is identifiable.	Problem and user are clearly defined with evidence.	Problem is well-bounded, evidence-based and compared with other ways of framing it.
Solution choice	You have a solution idea.	You explain why your approach fits the problem.	You compare AI and non-AI options and justify the simplest approach that works.
Prototype	A basic demonstration.	The prototype tests a core assumption.	The prototype is designed around measurable success and failure criteria.
Testing	You got feedback.	You recorded test evidence and made a change.	You ran structured tests, red-teamed failure modes and compared versions.
Responsible AI	You named a risk.	You identified privacy, bias, hallucination or oversight issues and a safeguard.	You analysed trade-offs, remaining risk, who's affected and who's accountable.
Presentation	You described the idea.	You communicated problem, prototype, test and learning clearly.	You used evidence, acknowledged uncertainty and handled challenge questions thoughtfully.

☆ Level up (optional challenge)

Add a simple evaluation plan with one quantitative measure (a number) and one qualitative measure (what people say). Run a second test cycle and compare the results with the first.

Badge earned: AI Innovator ✓ · Programme complete

Portfolio sheets

Copy these as often as you need. Each one matches a piece of evidence asked for in the chapters. Your teacher may give you printed versions.

Sheet A1 · Mission evidence sheet (any chapter)

Use this for any mission. Four things matter: what you predicted, what you tried, what actually happened, what you'd change.

Prompt	My response
Mission / chapter	
My prediction before I started	
What I tried	
Evidence / result	
What surprised me or failed	
What I changed or would change	
What I can now explain	
Where this matters outside the activity	

Sheet A2 · AI output verification log (Chapters 4, 5, 8)

One row per claim. Check against a source that is independent of the AI and of the original claim.

Claim	Source I checked	Supported / uncertain / wrong	What I changed

Sheet A3 · Model test log (Chapter 2)

At least ten rows. Then: accuracy = correct ÷ total. Put the same rows into a 2x2 confusion matrix.

#	Actual class	Prediction	Correct?	Observation (angle, light, background...)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Sheet A4 · Prompt experiment sheet (Chapter 4)

Change one thing at a time where you can, so you know what caused the difference.

Version	What I changed in the prompt	What changed in the output	Was it better? Why?
V1 – baseline			
V2			
V3			

Sheet A5 · Responsible data card (Chapters 3, 8)

Question	Notes
What data is collected?	
What is observed rather than typed in?	
What might be inferred?	
Why is it needed?	
What could go wrong?	
Who might be missing or misrepresented?	
What should be removed or minimised?	
What needs human review?	

Exit ticket (end of any session)

Five quick questions. Your honest answers help make the next version of this programme better.

Question	My answer
What did you discover today that you didn't know before?	
Where did you get stuck?	
What did you try when something didn't work?	
What part felt most useful or real?	
What would you change about this mission for the next group?	

Glossary

Plain meanings for the words in this book. Chapter key-word boxes give the short version; this is the fuller one.

AGI	Artificial General Intelligence. A hypothetical AI with broad, general capability across many kinds of task. Not the same as today's specialised systems.
Algorithm	A set of steps or rules for turning an input into an output.
AI model	A system trained or designed to recognise patterns, generate outputs or make predictions.
Automation bias	Trusting a machine's output more than the evidence justifies, just because a machine produced it.
Bias	A systematic skew that leads to unfair, unrepresentative or misleading results.
Class label	The category attached to an example in supervised learning, e.g. "cup".
Confusion matrix	A table comparing what was actually true with what the model predicted, so the different kinds of error can be seen.
Data minimisation	Collecting only the data genuinely needed for a stated purpose.
Deep learning	Machine learning that uses multi-layer neural networks. Mentioned in this book only as an idea.
Disclosure	Saying clearly when and how AI helped with a piece of work.
Generalisation	A model's ability to perform sensibly on examples it never saw during training.
Generative AI	AI that creates new content such as text, images, audio or code from learned patterns.
Hallucination	A fluent, confident output that is unsupported, wrong or made up.
Human-in-the-loop	A workflow where a person reviews, decides or approves at important points.
Inferred data	Information predicted about a person or situation from other data, rather than provided directly.
Large language model (LLM)	A generative model trained on huge amounts of text to predict and generate language.
Lateral reading	Checking a source by leaving it and finding out what other, independent sources say.
Machine learning	Methods where a system learns patterns from data instead of being given every rule explicitly.

Narrow AI	AI built for a specific task or family of tasks. All the AI in this book.
Prompt	The instructions and context given to a generative AI system.
Provenance	Where a piece of content came from and how it reached you.
Red-teaming	Deliberately trying to make your own system fail or misbehave, to find weaknesses before someone else does.
Shortcut learning	When a model relies on an easy, unintended clue (like background colour) instead of the feature you meant it to learn.
Test data	Examples used to evaluate a model after training. Ideally never reused from the training set.
Training data	Examples used to teach a machine-learning model.
Verification	Checking a claim against independent, reliable evidence.

About this programme

AI in Practice is a 30-hour Transition Year programme built around active, inquiry-based and self-directed learning. It is designed so that you learn AI by using it, testing it, questioning it and building with it. Passive instruction is kept deliberately short.

What you're working towards

The programme is aligned to the nine learning outcomes of the NCCA/CSLINC *Exploring Artificial Intelligence* Transition Year micro-module. Each chapter opener tells you which ones it links to.

Outcome	You should be able to...
LO1	Outline fundamental concepts of artificial intelligence.
LO2	Use and develop a basic AI model to perform tasks.
LO3	Evaluate AI models using standard metrics.
LO4	Examine the positive and negative potential of AI models.
LO5	Investigate how AI developers gather and collect data.
LO6	Examine ethical considerations around data used for AI.
LO7	Explore generative AI capabilities, particularly in an educational context.
LO8	Discuss possible AI futures, with emphasis on societal impacts.
LO9	Investigate misinformation and bias in AI use and ways to reduce their impact.

How the hours add up

Chapter	Hours	Main outcomes
1 · AI & Me	2	LO1, LO4
2 · Teaching a Machine	4	LO1, LO2, LO3, LO9
3 · Data Detective	3	LO5, LO6, LO9
4 · Generative AI & Prompting	4	LO7, LO9
5 · Trust, Bias & Misinformation	3	LO4, LO6, LO9
6 · AI for Learning & Work	3	LO7
7 · Our AI Future	3	LO4, LO8
8 · AI Innovation Project	8	LO1-LO9
Total	30	LO1-LO9

What this programme deliberately avoids

- Long slide-led lectures on AI terminology.
- Teaching prompting as “secret phrases” to memorise.
- Treating AI output as authoritative evidence.
- Requiring coding from every student.
- Using grades as the main motivator.
- Mistaking a polished prototype for deep learning.

Sources and alignment

NCCA / CSLINC, *Exploring Artificial Intelligence (AI)*, Transition Year micro-module, October 2025 (curriculumonline.ie). NCCA, *Transition Year Programme Statement* (ncca.ie). This is an independently designed pilot curriculum aligned to the themes and learning outcomes of the NCCA/CSLINC Exploring AI micro-module. It is not an official NCCA publication and should not be represented as such.

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